

Andrew Zheng

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EXPERIENCE

Software Engineer Intern | [Miovision](#) | Waterloo, ON

May 2026 – Aug 2026

- Re-engineered RAUC firmware update path for Yocto-based x86-64 Linux devices in Node.js and AWS IoT via adaptive block hash index HTTP streaming to reduce OTA bandwidth cost by **98%** (1.4GB to 21MB)
- Built end-to-end HIL release pipeline for embedded ARM64 Linux devices in Node.js and AWS/GitLab on Terraform-provisioned, Ansible-configured OTA runners to poll **15** health metrics over **48h** with **<1%** frame-drop
- Deployed 360° camera emulator app to embedded ARM64 Linux devices on Firecracker microVM in Go/Docker to virtualize H.264 RTSP streaming + camera API, enabling camera-free HIL development
- Architected fleet-wide dual lookup for device search engine in Kotlin/Spring Boot to stream schema registry Kafka data to Kubernetes-hosted index, eliminating **100%** of stale results across **43K+** devices

Embedded Systems Developer | [Waterloo Rocketry](#) | Waterloo, ON

Sep 2025 – Present

- Designed 4-layer recovery parachute PCB for the **world record highest amateur liquid rocket (63,497 ft)**, integrating dual 9V LiPo power stages with StratoLogger altimeters and Rocket CAN bus
- Engineered hardware fail-safe for PIC18 microcontroller to drive a P/N-channel MOSFET network to arm dual altimeters, fire pyros, and guarantee parachute deployment at peak altitude

Founder & Case Coach | [Case Tutoring Startup](#) | Toronto, ON

May 2024 – Aug 2025

- Drove **\$10K+** revenue through GTM campaign for self-founded business case contest coaching service, scaling across **150+** high school students from **10+** schools
- Won **1st Place** globally and coached **50+** qualifiers/**30+** international finalists at DECA ICDC, America's largest high school business case contest (230K+ members)

PROJECTS

Allocate | [macOS CPU/Thermal Fleet Controller](#) | *Rust, XNU, Mach IPC, Prometheus, Grafana*

[GitHub](#) 

- Built system-level macOS CPU governor using XNU QoS and POSIX policies to steer workloads across P/E cores and reduce background CPU consumption by **79%** with **100%** app uptime
- Engineered Rust FFI telemetry daemon to sample CPU, AppleSMC, IOReport, DVFS, and SoC power metrics every **500ms**, displayed on native AppKit live CPU chart
- Built 18-sensor Rust PID-based thermal controller reading CPU/GPU/ANE to dynamically adjust E-core policies through Mach/XPC, reducing CPU temperature by **12°C/22°F** without hardware fans
- Shipped native end-to-end SwiftUI app and Prometheus/Grafana fleet observability pipeline with configurable thresholds for per-app CPU controls, thermal, power, and throttling analysis

GitIRL Bot | [Hack the North 2026 Winner](#) | *Python, FastAPI, SLAM, OpenAI API*

[Devpost](#) 

- Won Best use of Bracket Bot Hardware at Canada's Biggest Hackathon by building an autonomous version control system to map world states and command a humanoid robot to restore changes
- Designed fleet orchestration pipeline across Bracket Bot and cloud platform, indexing **6.9M+** telemetry points in Elasticsearch to enable hybrid semantic search processing for robotic actions

CrawlStars | [Distributed Search Engine](#) | *Golang, JavaScript, MongoDB, REST API*

[GitHub](#) 

- Engineered concurrent crawlers in producer-consumer architecture to process **2000+** pages/min in **<15MB**
- Optimized deduplication to enable O(1) lookup for **50K+** URLs and filter **70%** of redundant crawls
- Deployed search engine to calculate SEO relevance via MongoDB Atlas aggregation pipeline for **50K+** URLs

SKILLS

Languages: C, C++, Python, Rust, Go, Java, Kotlin, Swift, JavaScript, TypeScript, HTML, CSS, Ruby, Verilog

Infrastructure: AWS, Linux, Docker, Kubernetes, Terraform, Ansible, Kafka, CI/CD, Git, Prometheus, Grafana

Backend: Node.js, Express.js, FastAPI, Spring Boot, Ruby on Rails, REST API, MongoDB, Jupyter

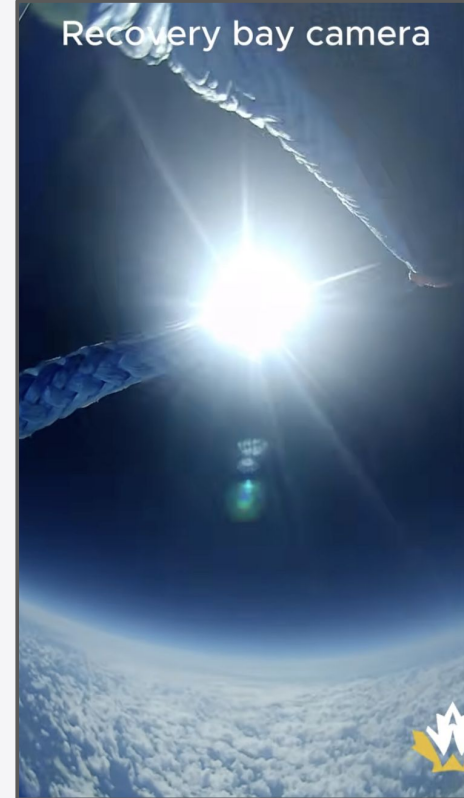
Frontend/Other: React, Next.js, Tailwind, Streamlit, XGBoost, SLAM, VLA, Ubuntu, Mach, Apple IOKit

EDUCATION

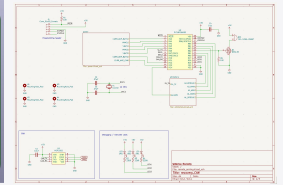
University of Waterloo | *BASc, Computer Engineering*

Sep 2025 – Apr 2030

Firmware: Embedded C (PIC18) | CAN Bus Telemetry Aggregation | ADC Monitoring



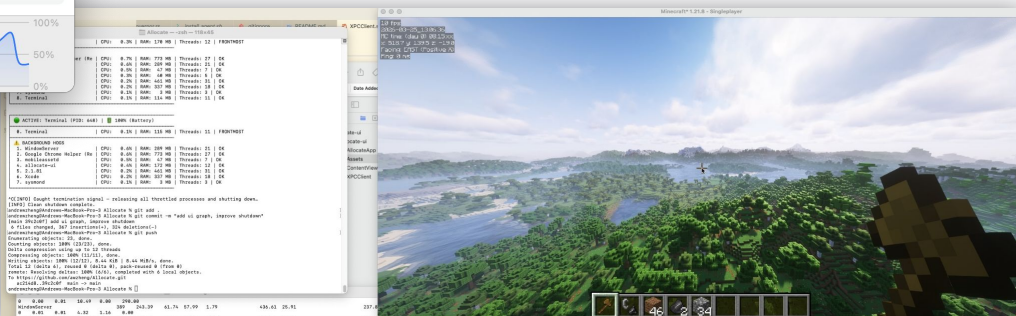
Recovery parachute in action



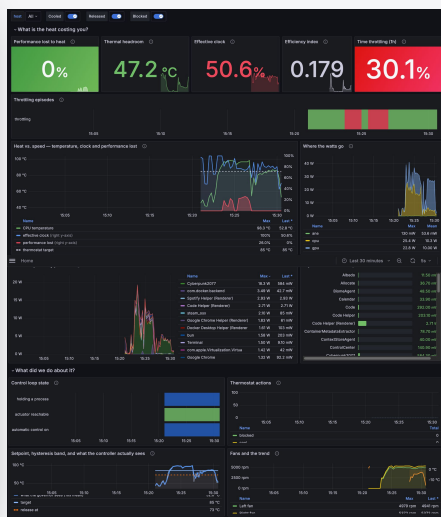
Python, FastAPI, React/Javascript, Docker, ESQ, OpenAI API | Humanoid VLA with 4D Memory



[\[GitHub\]](#) [\[Website\]](#)



AppleSMC, IOKit | Achieved 12°C/22°F Fanless CPU Cooling | 18 Local Hardware Sensors



```

graph TD
    subgraph Inputs
        direction TB
        A[Mac User]
        B[AppleSMC / IOKit]
        C[IOReport / libproc]
    end

    A -- "View and control" --> D[SwiftUI Menu Bar]
    A -- "Inspect and govern" --> E[Ratatui CLI]
    B -- "Sensor readings" --> F[Rust Thermal Engine]
    C -- "Power and processes" --> F

    D -- "C ABI snapshots" --> F
    E -- "Native Rust calls" --> F

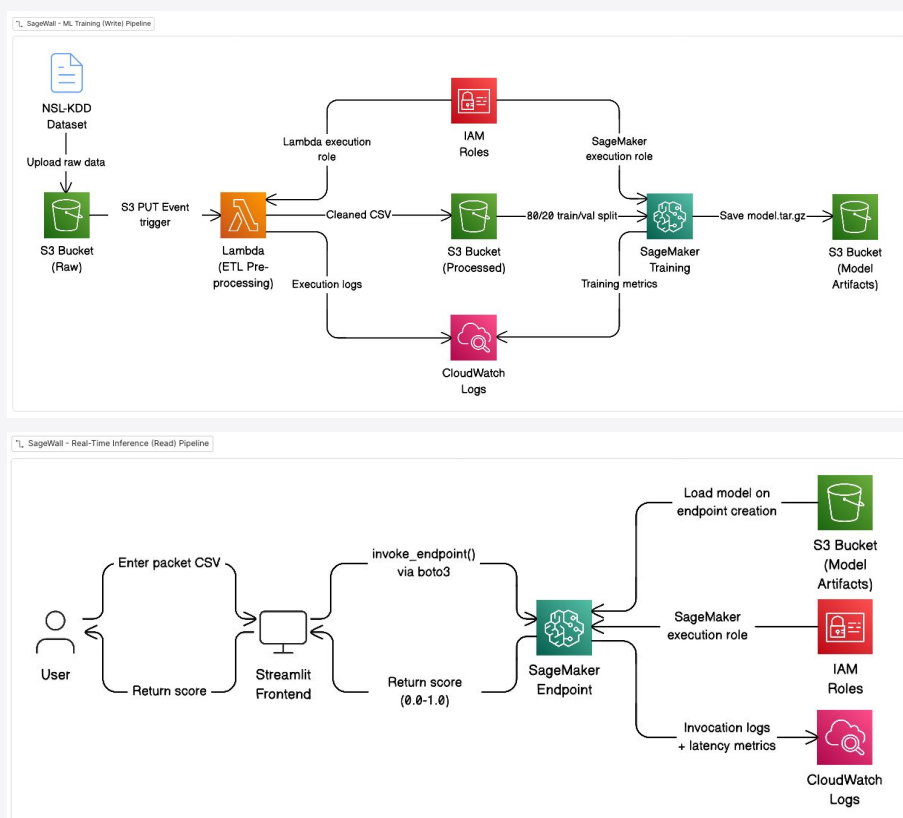
    F -- "Expose derived metrics" --> G[Metrics HTTP API]
    F -- "Request E-core hold" --> H[Allocate XPC Daemon]

    G -- "Scrape host metrics" --> I[Grafana Alloy]
    G -- "Local scrape" --> J[(Prometheus TSDB)]
    I -- "Remote write" --> J

    H -- "Apply QoS policy" --> K[Apple Silicon Scheduler]

    J -- "Fleet visualization" --> L[Grafana Dashboards]
  
```

The diagram illustrates the architecture of the Rust Thermal Engine. It shows how data from various system components (Mac User, AppleSMC / IOKit, IOReport / libproc) is processed by the Rust Thermal Engine. The engine then exposes derived metrics via a Metrics HTTP API and requests E-core holds via an Allocate XPC Daemon. The Metrics HTTP API scrapes host metrics and local scrapes, which are then processed by Grafana Alloy and Prometheus TSDB. The Allocate XPC Daemon applies QoS policy to the Apple Silicon Scheduler. Finally, the Prometheus TSDB provides fleet visualization to Grafana Dashboards.



Golang, MongoDB | O(1) lookup time | 2000+ URLs/minute | <15MB memory usage

